

RADAR LEVEL TRANSMITTER

TAGS: LT-01

Technical Information Micropilot FMR20 HART

Free space radar

Level measurement for liquids



Application

- Ingress protection: IP66/68 / NEMA 4X/6P
- Measuring range: up to 20 m (66 ft)
- Process temperature: -40 to 80 °C (-40 to 176 °F)
- Process pressure: -1 to 3 bar (-14 to 43 psi)
- Accuracy: up to ± 2 mm (0.08 in)
- International explosion protection certificates

Your benefits

- Level measurement for liquids in storage tanks, open basins, pump shafts and canal systems
- Radar measuring device with *Bluetooth*® wireless technology and HART communication
- Simple, safe and secure wireless remote access – ideal for installation in hazardous areas or places difficult to reach
- Commissioning, operation and maintenance via free iOS / Android app SmartBlue – saves time and reduces costs
- Full PVDF body - for a long sensor lifetime
- Hermetically sealed wiring and fully potted electronics – eliminates water ingress and allows operation under harsh environmental conditions
- Most compact radar due to unique radar chip design – fits in limited space installations
- Best price-performance-ratio radar

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Important document information

Symbols used

Symbols for certain types of information and graphics

-  **Permitted**
Procedures, processes or actions that are permitted
-  **Forbidden**
Procedures, processes or actions that are forbidden
-  **Tip**
Indicates additional information
-  **Reference to documentation**
-  **Reference to graphic**
-  **Notice or individual step to be observed**
-  **Series of steps**
-  **Result of a step**
-  **Item numbers**
-  **Views**

Terms and abbreviations

BA

Document type "Operating Instructions"

KA

Document type "Brief Operating Instructions"

TI

Document type "Technical Information"

SD

Document type "Special Documentation"

XA

Document type "Safety Instructions"

PN

Nominal pressure

MWP

MWP (Maximum working pressure/max. process pressure)

The MWP can also be found on the nameplate.

ToF

Time of Flight

FieldCare

Scalable software tool for device configuration and integrated plant asset management solutions

DeviceCare

Universal configuration software for Endress+Hauser HART, PROFIBUS, FOUNDATION Fieldbus and Ethernet field devices

DTM

Device Type Manager

 ϵ_r (Dk value)

Relative dielectric constant

Operating tool

The term "operating tool" is used in place of the following operating software:

- FieldCare / DeviceCare, for operation via HART communication and PC
- SmartBlue (app), for operation using an Android or iOS smartphone or tablet

BD

Blocking Distance; no signals are analyzed within the BD.

PLC

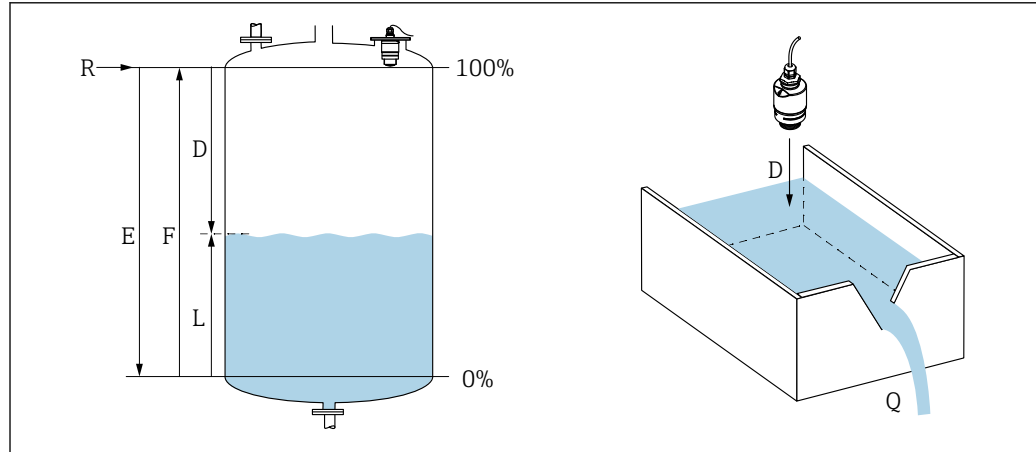
Programmable logic controller (PLC)

Product life cycle

Engineering	■ Proven radar measuring technology ■ Level measurement and open channel flow measurement for Ex and non-Ex areas ■ Flooding detection ■ Wide range of installation possibilities and accessories ■ Highest degree of ingress protection ■ 2D/3D drawings ■ Spec Sheet Producer ■ Applicator Selection tool for selecting the perfect measurement solution  Device not compatible with transmitters and sensors that use ultrasonic measurement technology (e.g. Prosonic FMU9x, FDU9x)
Procurement	■ Best price-performance-ratio radar ■ Global availability ■ Order code includes variety of mounting accessories and decentralized RIA15 process indicator for HART
Installation	■ Rear and front thread for flexible installation ■ Slip-on flange for nozzle installation ■ Complete measuring point: including mounting accessory, RIA15 and flooding protection tube
Commissioning	■ Quick and easy setup with the SmartBlue app and DeviceCare / FieldCare or RIA15 ■ No additional tools or adapters required ■ Local languages (up to 15)
Operation	■ Continuous self-monitoring ■ Diagnostics information according to NAMUR NE107 with remedial measures in the form of plain text messages ■ Signal curve via SmartBlue (app) and DeviceCare / FieldCare ■ Encrypted single point-to-point data transmission (tested by Fraunhofer Institute) and password-protected communication via <i>Bluetooth®</i> wireless technology
Maintenance	■ No maintenance required ■ Technical experts on-call around the world
Retirement	■ Environmentally responsible recycling concepts ■ RoHS compliance (restriction of certain hazardous substances), lead-free soldering of electronic components

Measuring principle

The Micropilot is a "downward-looking" measuring system, which functions according to the time-of-flight (ToF) method. It measures the distance from the reference point **R** to the product surface. Radar pulses are emitted by an antenna, reflected off the product surface and received again by the radar system.



1 Level / flow calibration parameter

E Empty calibration (= zero)

F Full calibration (= span)

D Measured distance

L Level ($L = E - D$)

Q Flow rate at measuring weirs or channels (calculated from the level using linearization)

R Reference point

Input

The reflected radar pulses are received by the antenna and transmitted to the electronics. A microprocessor evaluates the signals and identifies the level echo caused by the reflection of the radar pulses at the product surface. This clear signal detection system benefits from over 30 years' experience with time-of-flight procedures.

The distance **D** to the product surface is proportional to the time of flight **t** of the pulse:

$$D = c \cdot t / 2,$$

where **c** is the speed of light.

Based on the known empty distance **E**, the level **L** is calculated:

$$L = E - D$$

Output

The Micropilot is calibrated by entering the empty distance **E** (= zero point) and the full distance **F** (= span).

- Current output: 4 to 20 mA
- Digital output (HART, SmartBlue): 0 to 15 m (0 to 49 ft)¹⁾ or 0 to 20 m (0 to 66 ft) depending on antenna version

1) order code 010 (approval) "GA", "GB", "GR": 0 to 10 m (0 to 32.8 ft)

Input

Measured variable	The measured variable is the distance between the reference point and the product surface. The level is calculated based on E , the empty distance entered.
Measuring range	Maximum measuring range → ■ Device with 40 mm (1.5 in) antenna: 15 m (49 ft) ²⁾ ■ Device with 80 mm (3 in) antenna: 20 m (66 ft) Installation requirements ■ Recommended tank height > 1.5 m (5 ft) for media with low ϵ_r value ■ Open channel minimum width 0.5 m (1.6 ft) ■ Calm surfaces ■ No agitators ■ No buildup ■ Relative dielectric constant $\epsilon_r > 4$ Contact Endress+Hauser for lower ϵ_r values Usable measuring range The usable measuring range depends on the antenna size, the medium's reflective properties, the installation position and any possible interference reflections. Media groups ■ $\epsilon_r = 4$ to 10 e.g. concentrated acid, organic solvents, ester, aniline, alcohol, acetone. ■ $\epsilon_r > 10$ e.g. conductive liquids, aqueous solutions, diluted acids and bases Reduction of the maximum possible measuring range by: ■ Media with bad reflective properties (= low ϵ_r value) ■ Formation of buildup, particularly of moist products ■ Strong condensation ■ Foam generation ■ Freezing of sensor
Operating frequency	K-band (~ 26 GHz)
Transmission power	Mean power density in the direction of the beam ■ At a distance of 1 m (3.3 ft): < 12 nW/cm² ■ At a distance of 5 m (16 ft): < 0.4 nW/cm²

Output

Output signal	4 to 20 mA An 4 to 20 mA interface is used for measured value output and to power to the device.
Digital output	HART® ■ Signal encoding; FSK ± 0.5 mA over current signal ■ Data transmission rate; 1 200 Bit/s Bluetooth® wireless technology (available as an optional extra) The device has a <i>Bluetooth®</i> wireless technology interface and can be operated and configured via this interface using the SmartBlue app.

2) order code 010 (approval) "GA", "GB", "GR": 10 m (32.8 ft)

- The range under reference conditions is 25 m (82 ft)
- Incorrect operation by unauthorized persons is prevented by means of encrypted communication and password encryption
- The *Bluetooth*® wireless technology interface can be deactivated

Signal on alarm

Depending on the interface, failure information is displayed as follows:

- Current output
Alarm current: 22.5 mA (in accordance with NAMUR recommendation NE 43)
- Operating tool via digital communication (HART) or SmartBlue (app)
 - Status signal (as per NAMUR Recommendation NE 107)
 - Plain text display with remedial action

Linearization

The linearization function of the device allows the conversion of the measured value into any unit of length, weight, flow or volume. In DeviceCare and FieldCare, there are preprogrammed linearization tables for volume calculation in vessels.

Pre-programmed linearization curves

- Horizontal cylindrical tank
- Sphere
- Tank with pyramid bottom
- Tank with conical bottom
- Tank with flat bottom

Other linearization tables of up to 32 value pairs can be entered manually.

Protocol-specific data, HART

Manufacturer ID

17 (0x11)

Device type ID

44 (0x112c)

HART specification

7.0

Device description files (DTM)

Information and files under:

- www.endress.com
- www.hartcomm.org

HART load

Min. 250 Ω

HART device variables

Assignment of HART device variables is fixed and cannot be changed.

- **Measured values for PV (primary variable)**
Level linearized
- **Advanced diag. measured values for SV (secondary variable)**
Distance
- **Advanced diag. measured values for TV (tertiary variable)**
Relative echo amplitude
- **Advanced diag. measured values for QV (quarternary variable)**
Temperature

Supported functions

Additional transmitter status

Multidrop current

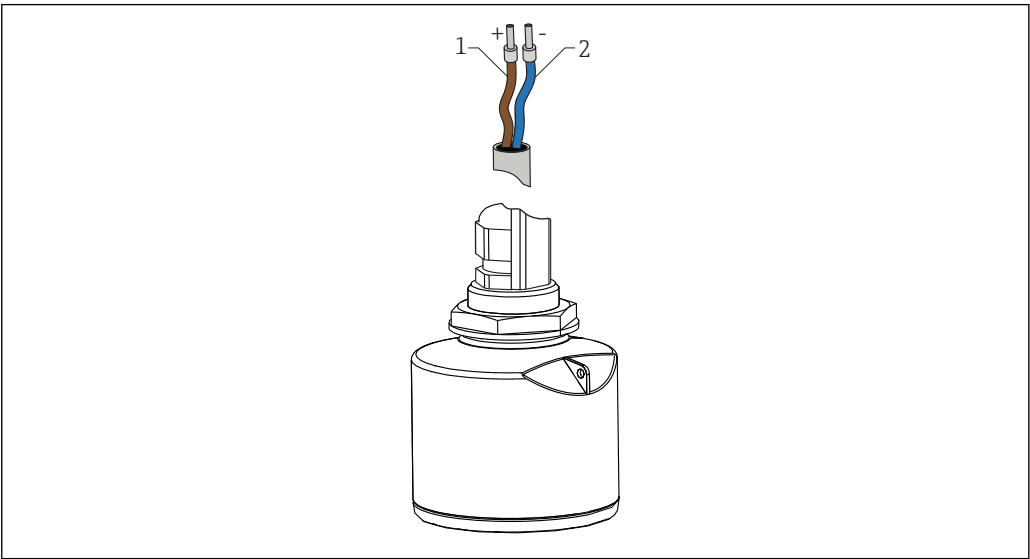
4 mA

Time for connection setup

< 1 s

Electrical connection

Cable assignment

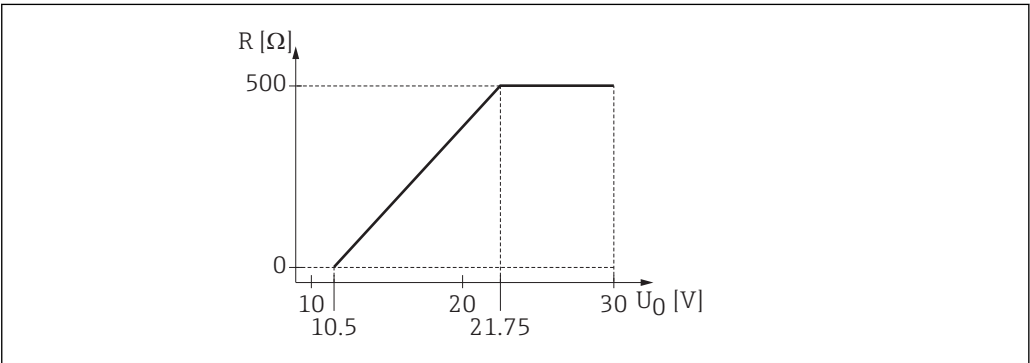


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- 2 Cable assignment
- 1 Plus, brown wire
 - 2 Minus, blue wire

Supply voltage

10.5 to 30 V_{DC}
An external power supply is necessary.



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- 3 Maximum load R, depending on supply voltage U₀ of power supply unit

Battery operation

The sensor's *Bluetooth*® wireless technology communication can be disabled to increase the operating life of the battery.

Potential equalization

No special measures for potential equalization are required.



Various power supply units can be ordered as an accessory from Endress+Hauser.

Power consumption

Maximum input power: 675 mW

Current consumption

- Maximum input current: <25 mA
- Maximum start-up current: 3.6 mA

Starting time

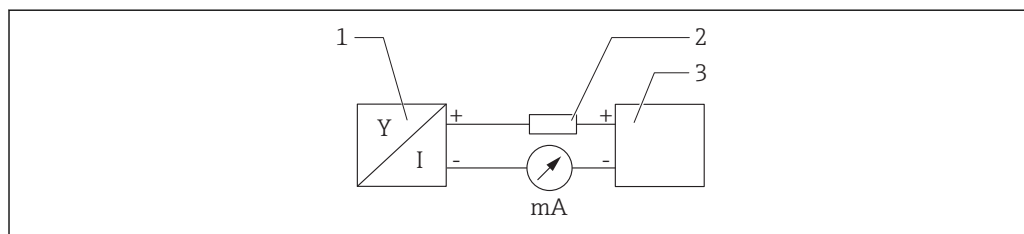
First stable measured value after 20 s (with supply voltage = 24 V_{DC})

Power supply failure

The configuration remains stored in the sensor.

Connecting the device**4 to 20 mA HART block diagram**

Connection of the device with HART communication, power source and 4 to 20 mA display



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 4 Block diagram of HART connection

1 Device with HART communication

2 HART resistor

3 Power supply



The HART communication resistor of $250\ \Omega$ in the signal line is always necessary in the case of a low-impedance power supply.

The voltage drop to be taken into account is:

Max. 6 V for $250\ \Omega$ communication resistor

HART device block diagram, connection with RIA15

FMR20 with RIA15 (incl. option for FMR20 basic configuration)



The RIA15 remote indicator can be ordered together with the device.

Product structure, feature 620 "Accessory enclosed":

- Option R4 "Remote indicator RIA15 non-hazardous area, field housing"
- Option R5 "Remote indicator RIA15 with explosion protection approval, field housing"



Alternatively available as an accessory, for details see Technical Information TI01043K and Operating Instructions BA01170K

Terminal assignment RIA15

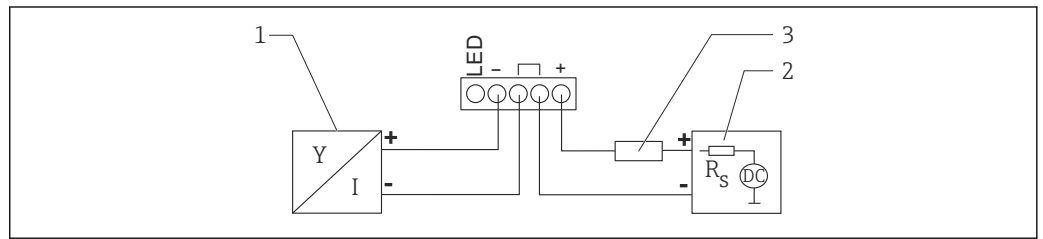
- +
Positive connection, current measurement
- -
Negative connection, current measurement (without backlighting)
- LED
Negative connection, current measurement (with backlighting)
- $\perp$
Functional grounding: Terminal in housing



The RIA15 process indicator is loop-powered and does not require any external power supply.

The voltage drop to be taken into account is:

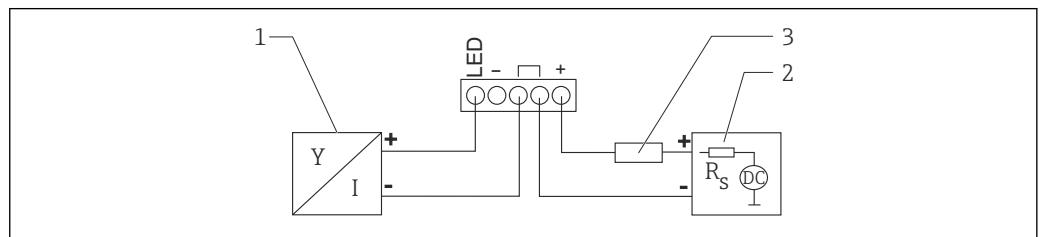
- ≤ 1 V in the standard version with 4 to 20 mA communication
- ≤ 1.9 V with HART communication
- and an additional 2.9 V if display light is used

Connection of the HART device and RIA15 without backlighting

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5 Block diagram of HART device with RIA15 process indicator without light

- 1 Device with HART communication
- 2 Power supply
- 3 HART resistor

Connection of the HART device and RIA15 with backlighting

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6 Block diagram of HART device with RIA15 process indicator with light

- 1 Device with HART communication
- 2 Power supply
- 3 HART resistor

Block diagram of HART device, RIA15 with installed HART communication resistor module

 The HART communication module for installation in the RIA15 can be ordered together with the device.

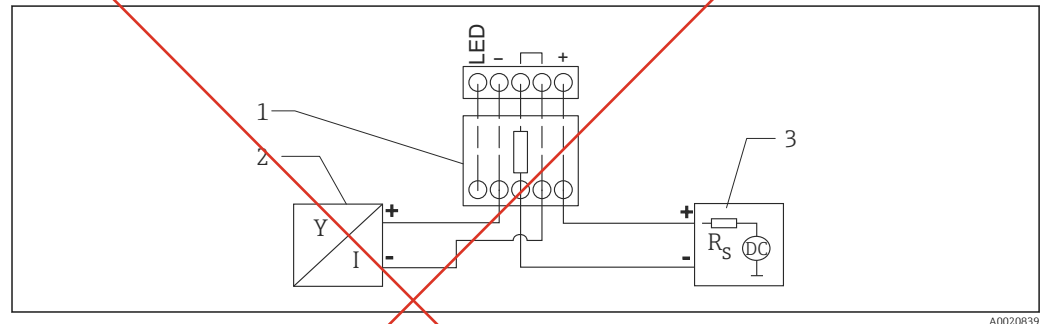
Product structure, feature 620 "Accessory enclosed":

Option R6 "HART communication resistor hazardous / non-hazardous area"

The voltage drop to be taken into account is:

Max. 7 V

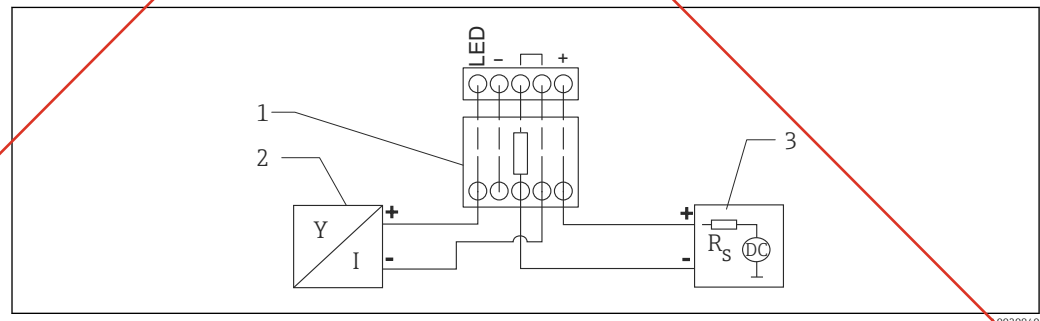
 Alternatively available as an accessory, for details see Technical Information TI01043K and Operating Instructions BA01170K

Connection of the HART communication resistor module, RIA15 without backlighting

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 7 Block diagram of HART device, RIA15 without light, HART communication resistor module

- 1 HART communication resistor module
- 2 Device with HART communication
- 3 Power supply

Connection of the HART communication resistor module, RIA15 with backlighting

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 8 Block diagram of HART device, RIA15 with light, HART communication resistor module

- 1 HART communication resistor module
- 2 Device with HART communication
- 3 Power supply

Cable specification

Unshielded cable, wire cross-section 0.75 mm²

- Resistant to UV and weather conditions as per ISO 4892-2
- Flame resistance according to IEC 60332-1-2

As per IEC/EN 60079-11 section 10.9, the cable is designed for a tensile strength of 30 N (6.74 lbf) (over a period of 1 h).

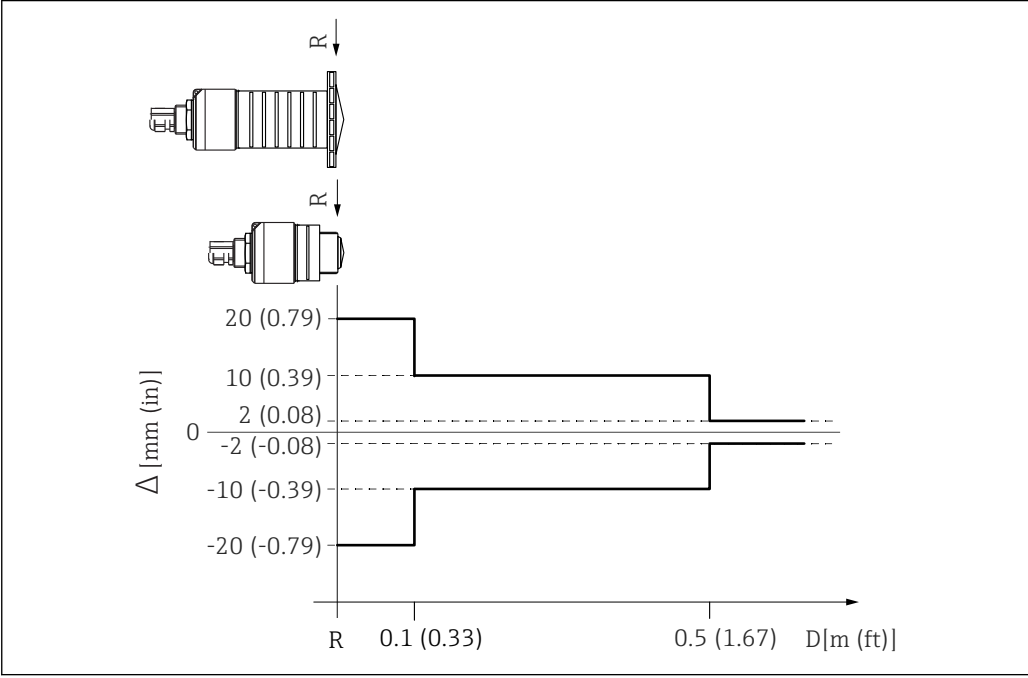
The device is supplied with 5 m (16 ft) cable length as standard. Cable lengths 10 m (33 ft) and 20 m (66 ft) are optionally available.

Lengths can be selected by the user up to an overall length of 300 m (980 ft) and are available by the meter (order option "8") or foot (order option "A").

Overvoltage protection

The device is equipped with integrated overvoltage protection.

Performance characteristics

Reference operating conditions	- Temperature = +24 °C (+75 °F) ±5 °C (±9 °F) - Pressure = 960 mbar abs. (14 psia) ±100 mbar (±1.45 psi) - Humidity = 60 % ±15 % - Reflector: metal plate with diameter ≥ 1 m (40 in) - No major interference reflections inside the signal beam
Maximum measured error	Typical data under reference operating conditions: DIN EN 61298-2, percentage values in relation to the span. Output, digital (HART, SmartBlue (app)) - Sum of non-linearity, non-repeatability and hysteresis: ±2 mm (±0.08 in) - Offset/zero point: ±4 mm (±0.16 in) Output, analog Only relevant for 4-20mA current output; add error of the analog value to the digital value - Sum of non-linearity, non-repeatability and hysteresis: ±0.02 % - Offset/zero point: ±0.03 % Differing values in near-range applications  9 Maximum measured error in near-range applications; values for standard version Δ Maximum measured error R Reference point of the distance measurement D Distance from reference point of antenna
Measured value resolution	Dead band as per EN61298-2: - Digital: 1 mm (0.04 in) - Analog: 4 μA
Response time	The response time can be configured. The following step response times apply (in accordance with DIN EN 61298-2) when damping is switched off: Tank height <20 m (66 ft) Sampling rate 1 s⁻¹

Response time

<3 s

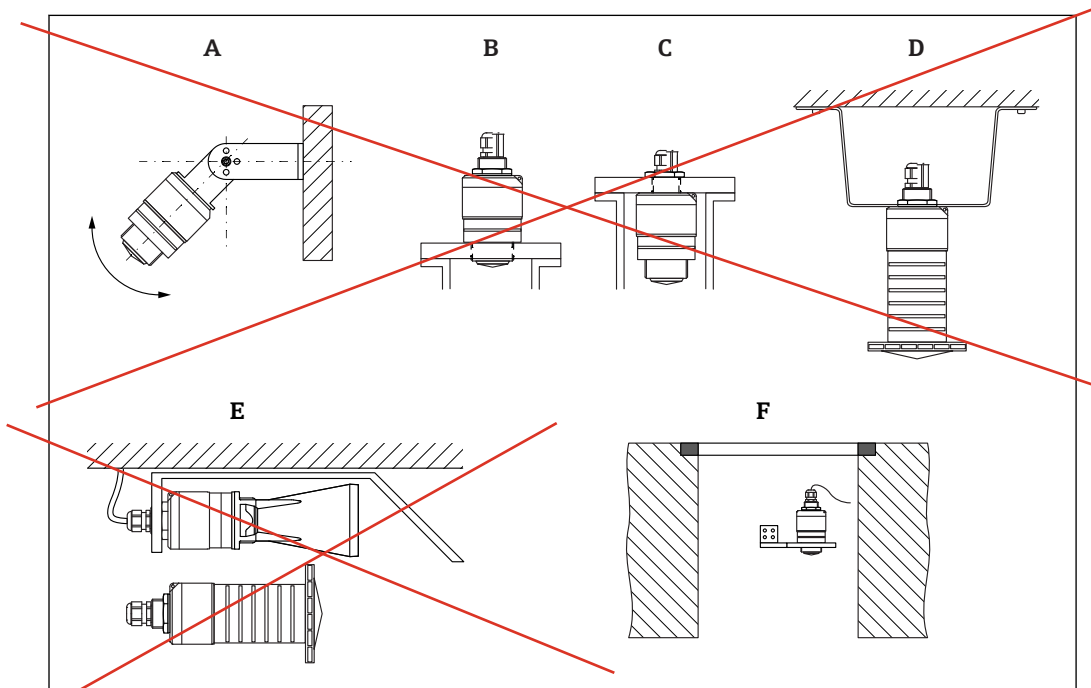


In accordance with DIN EN 61298-2, the step response time is the time following an abrupt change in the input signal up until the changed output signal has adopted 90% of the steady-state value for the first time.

Influence of ambient temperature**The measurements are carried out in accordance with EN 61298-3**

- Digital (HART, *Bluetooth*® wireless technology):
Standard version: average $T_C = \pm 3 \text{ mm } (\pm 0.12 \text{ in})/10 \text{ K}$
- Analog (current output):
 - Zero point (4 mA): average $T_K = 0.02 \text{ } \%/10 \text{ K}$
 - Span (20 mA): average $T_K = 0.05 \text{ } \%/10 \text{ K}$

Installation

Installation conditions**Installation types**

A0030605

10 Wall, ceiling or nozzle installation

- A Wall or ceiling mount, adjustable
- B Mounted at front thread
- C Mounted at rear thread
- D Ceiling installation with counter nut (included in delivery)
- E Horizontal installation in cramped spaces (wastewater shaft), 40 mm (1.5 in) antenna with flooding protection tube (accessory) or 80 mm (3 in) antenna recommended
- F Shaft wall mounting

**Caution!**

- The sensor cables are not designed as supporting cables. Do not use them for suspension purposes.
- Always operate the device in a vertical position in free-space applications.

Please note the limits for the diameter and length of the nozzle.

80 mm (3 in) antenna, installation inside nozzle

- D: min. 120 mm (4.72 in)
- L: max. 205 mm (8.07 in) + D × 4.5

80 mm (3 in) antenna, installation outside nozzle

- D: min. 80 mm (3 in)
- L: max. D × 4.5

40 mm (1.5 in) antenna, installation outside nozzle

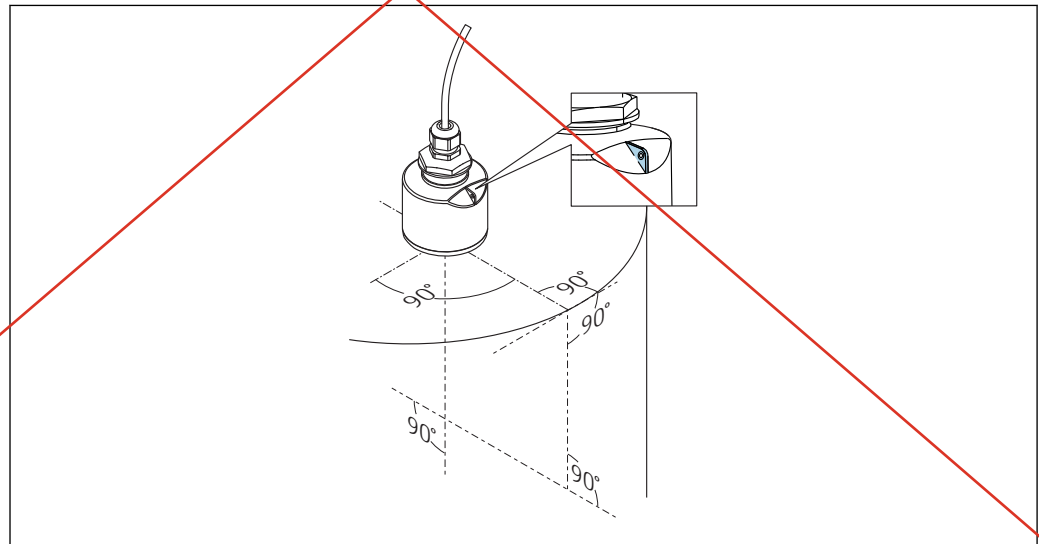
- D: min. 40 mm (1.5 in)
- L: max. D × 1.5

40 mm (1.5 in) antenna, installation inside nozzle

- D: min. 80 mm (3 in)
- L: max. 140 mm (5.5 in) + D × 1.5

Device alignment for installation on a vessel

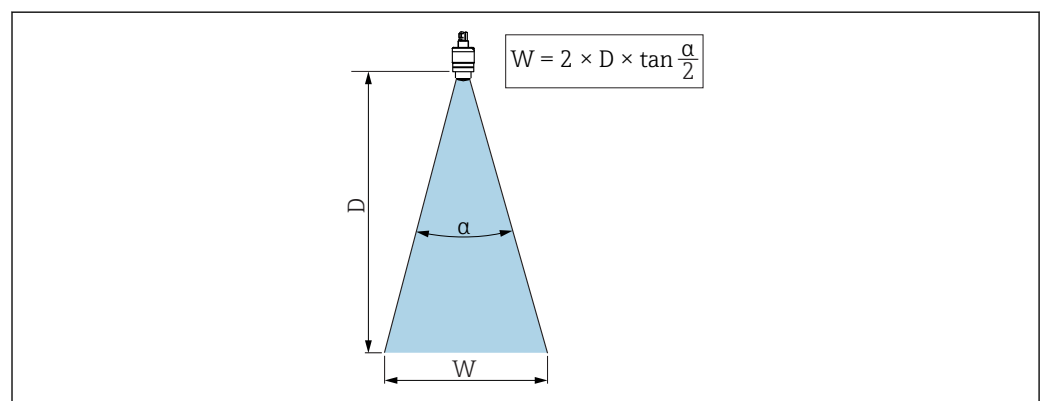
- Align the antenna vertically to the product surface.
- Align the eyelet with lug towards the vessel wall as well as possible.



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13 Device alignment for installation on a vessel

Beam angle



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14 Relationship between beam angle α , distance D and beamwidth diameter W

The beam angle is defined as the angle α , at which the power density of the radar waves reaches half the value of the maximum power density (3 dB width). Microwaves are also emitted outside the signal beam and can be reflected off interfering installations.

Beam diameter W as a function of beam angle α and distance D .

40 mm (1.5 in) antenna, $\alpha 30^\circ$

$$W = D \times 0.54$$

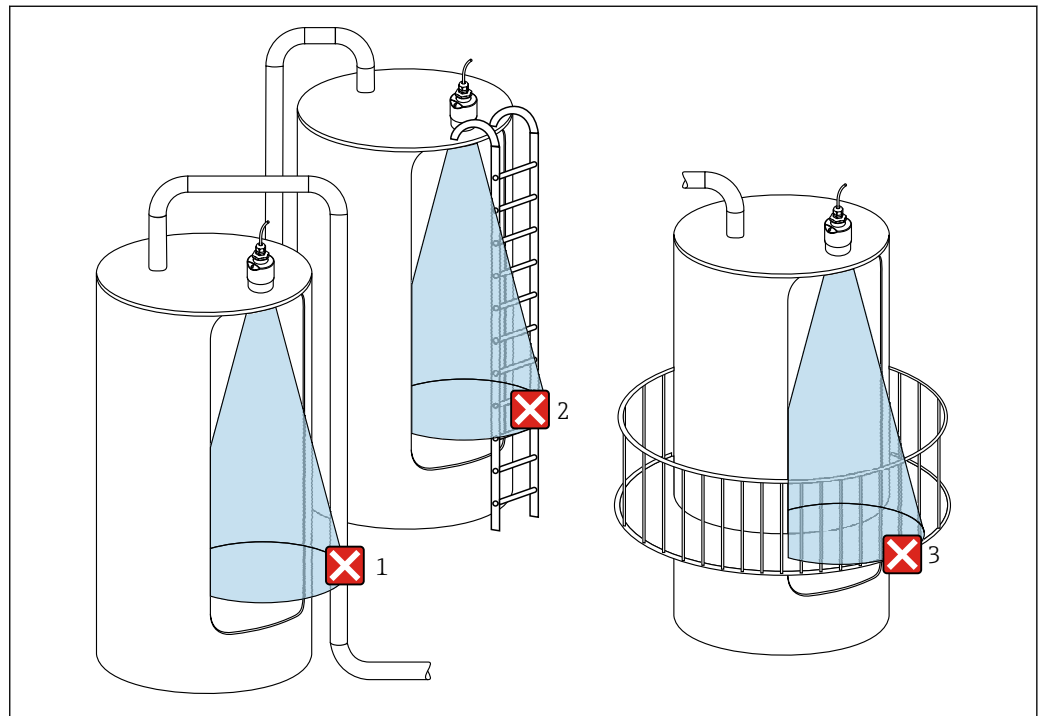
~~**40 mm (1.5 in) antenna with flooding protection tube, $\alpha 12^\circ$**~~

~~$$W = D \times 0.21$$~~

~~**80 mm (3 in) antenna with or without flooding protection tube, $\alpha 12^\circ$**~~

~~$$W = D \times 0.21$$~~

Measurement in plastic vessels



15 Measurement in a plastic vessel with a metallic, interfering installation outside of the vessel

- 1 Pipe, tubing
- 2 Ladder
- 3 Grate, railing

If the outer wall of the vessel is made of a non-conductive material (e.g. GFR), microwaves can also be reflected by interfering installations outside of the vessel.

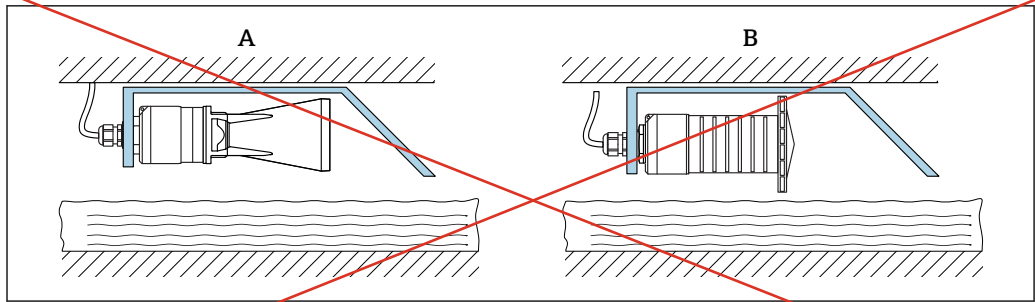
Please ensure there are no interfering installations made of a conductive material in the signal beam (see the beam angle section for information on calculating the beamwidth diameter).

Please contact the manufacturer for further information.

~~Protective hood~~

~~For outdoor use, a protective hood is recommended.~~

~~The protective hood can be ordered as an accessory or together with the device via the product structure "Accessory enclosed".~~

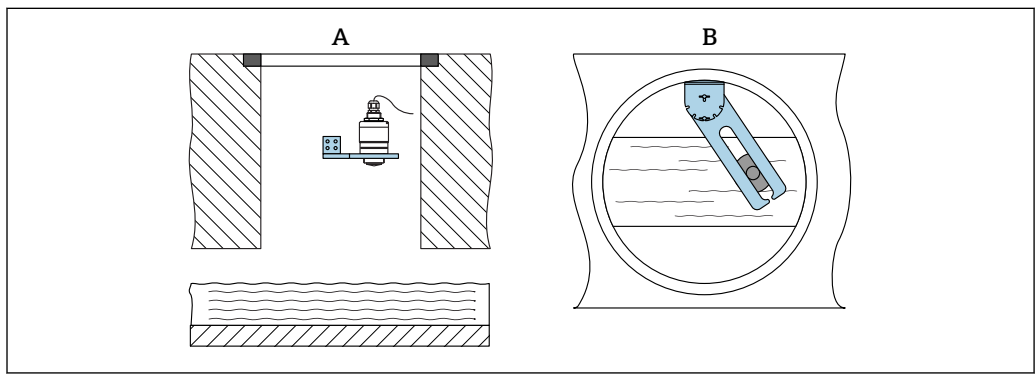


20 Installation of horizontal mounting bracket for sewer shafts

- A 40 mm (1.5 in) Antenna, with flooding protection tube (accessory)
B 80 mm (3 in) Antenna, without flooding protection tube

Mounting in a shaft

The pivoted mounting bracket is available as an accessory.



21 Mounting in a shaft, pivotable and adjustable

- A Arm with wall bracket
B Pivotable and adjustable arm (e.g. to align the device with the center of a channel)

Environment

Ambient temperature range

Measuring device: -40 to +80 °C (-40 to +176 °F)

i It may not be possible to use the Bluetooth connection at ambient temperatures > 60 °C (140 °F).

Outdoor operation in strong sunlight:

- Mount the device in the shade.
- Avoid direct sunlight, particularly in warm climatic regions.
- Use a weather protection cover.

Storage temperature

-40 to +80 °C (-40 to +176 °F)

Climate class

DIN EN 60068-2-38 (test Z/AD)

Operating altitude as per IEC 61010-1 Ed.3

Generally up to 2 000 m (6 600 ft) above sea level.

Degree of protection

Tested acc. to:

- IP66, NEMA 4X
- IP68, NEMA 6P (24 h at 1.83 m (6.00 ft) 1.83 m under water)

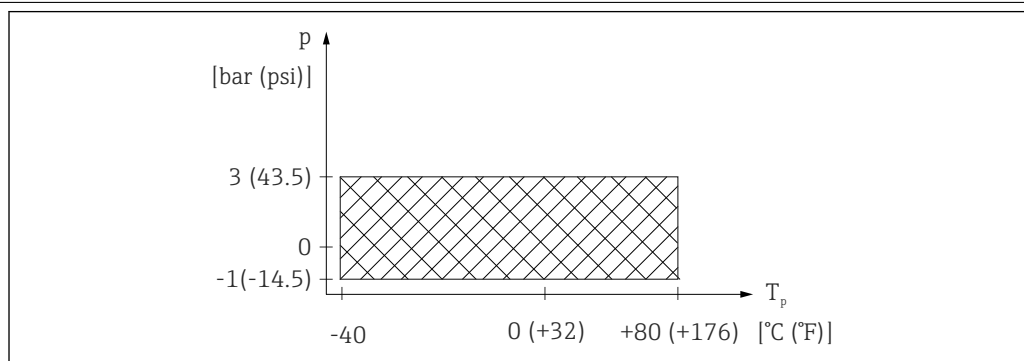
Vibration resistance

DIN EN 60068-2-64/IEC 60068-2-64: 20 to 2 000 Hz, 1 (m/s²)²/Hz

Electromagnetic compatibility (EMC)	Electromagnetic compatibility in accordance with all of the relevant requirements outlined in the EN 61000 series and NAMUR Recommendation EMC (NE 21). Details are provided in the Declaration of Conformity (www.endress.com/downloads).
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Process

Process temperature, process pressure



A0029007-EN

22 FMR20: Permitted range for process temperature and process pressure

Process temperature range

-40 to +80 °C (-40 to +176 °F)

Process pressure range, threaded process connection

- $p_{rel} = -1$ to 3 bar (-14.5 to 43.5 psi)
- $p_{abs} < 4$ bar (58 psi)

~~Process pressure range, UNI flange process connection~~

- ~~▪ $p_{rel} = -1$ to 1 bar (-14.5 to 14.5 psi)~~
- ~~▪ $p_{abs} < 2$ bar (29 psi)~~



The pressure range may be further restricted in the event of a CRN approval.

Dielectric constant

For liquids

- $\epsilon_r \geq 4$
- Contact Endress+Hauser for lower ϵ_r values



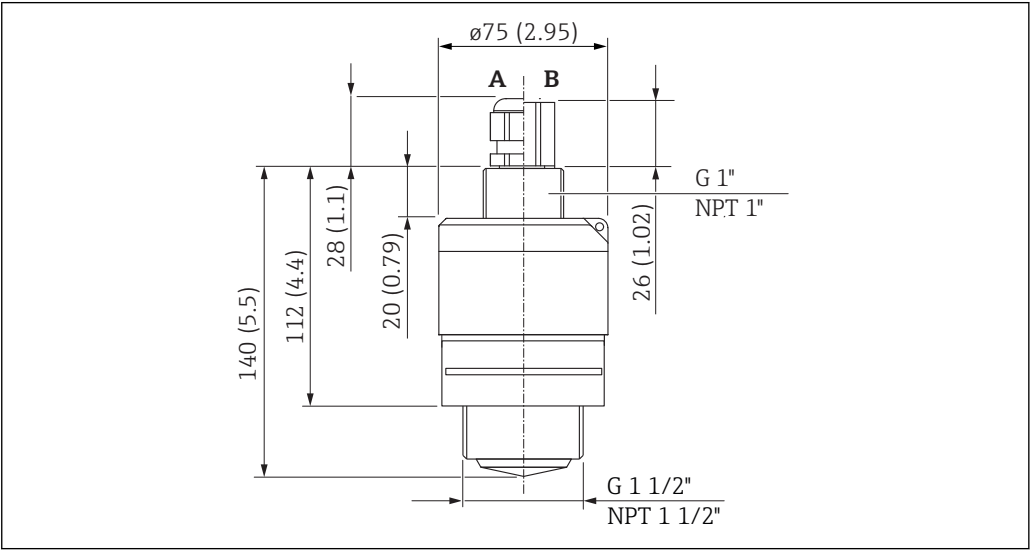
For dielectric constants (DC values) of many media commonly used in various industries refer to:

- the Endress+Hauser DC manual (CP01076F)
- the Endress+Hauser "DC Values App" (available for Android and iOS)

Mechanical construction

Dimensions

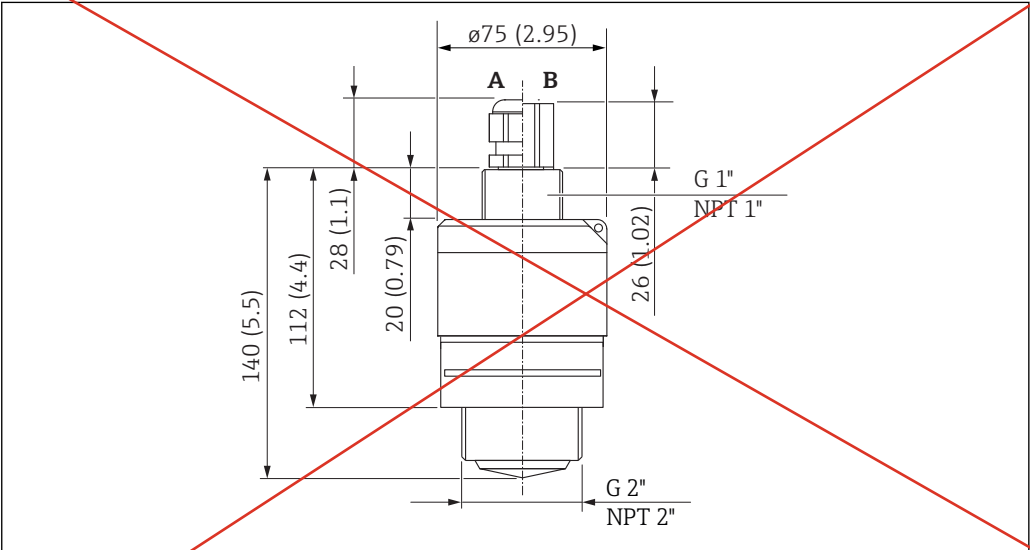
40 mm (1.5 in) Antenna with G 1-½" or MNPT 1-½" thread



23 Dimensions of G 1-½" or MNPT 1-½" process connection thread, engineering unit: mm (in)

- A Cable gland
- B FNPT ½" conduit

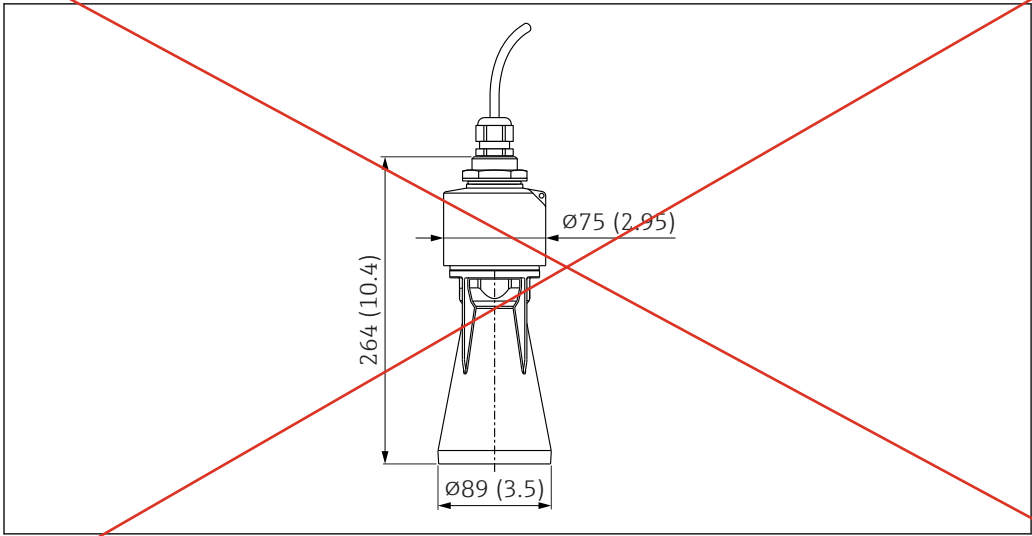
40 mm (1.5 in) Antenna with G 2" or MNPT 2" thread



24 Dimensions of G 2" or MNPT 2" process connection thread, engineering unit: mm (in)

- A Cable gland
- B FNPT ½" conduit

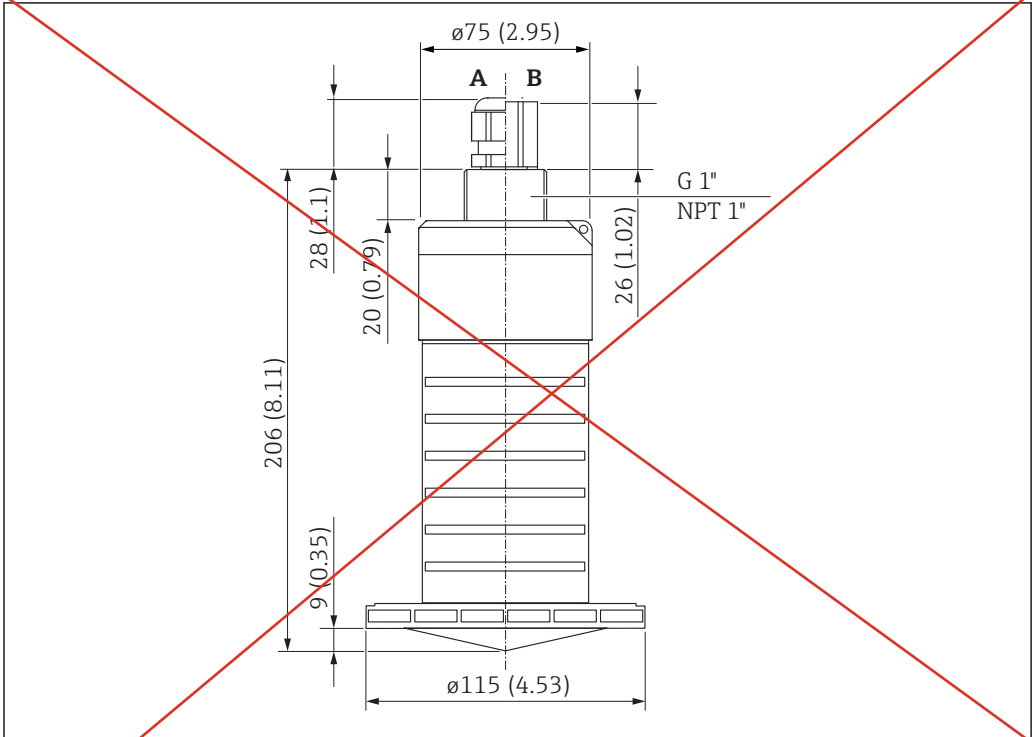
40 mm (1.5 in) antenna with flooding protection tube



25 Dimensions of 40 mm (1.5 in) antenna mounted with flooding protection tube, engineering unit: mm (in)

The flooding protection tube, metalized PBT-PC, can be ordered together with the device via the product structure "Accessory enclosed".

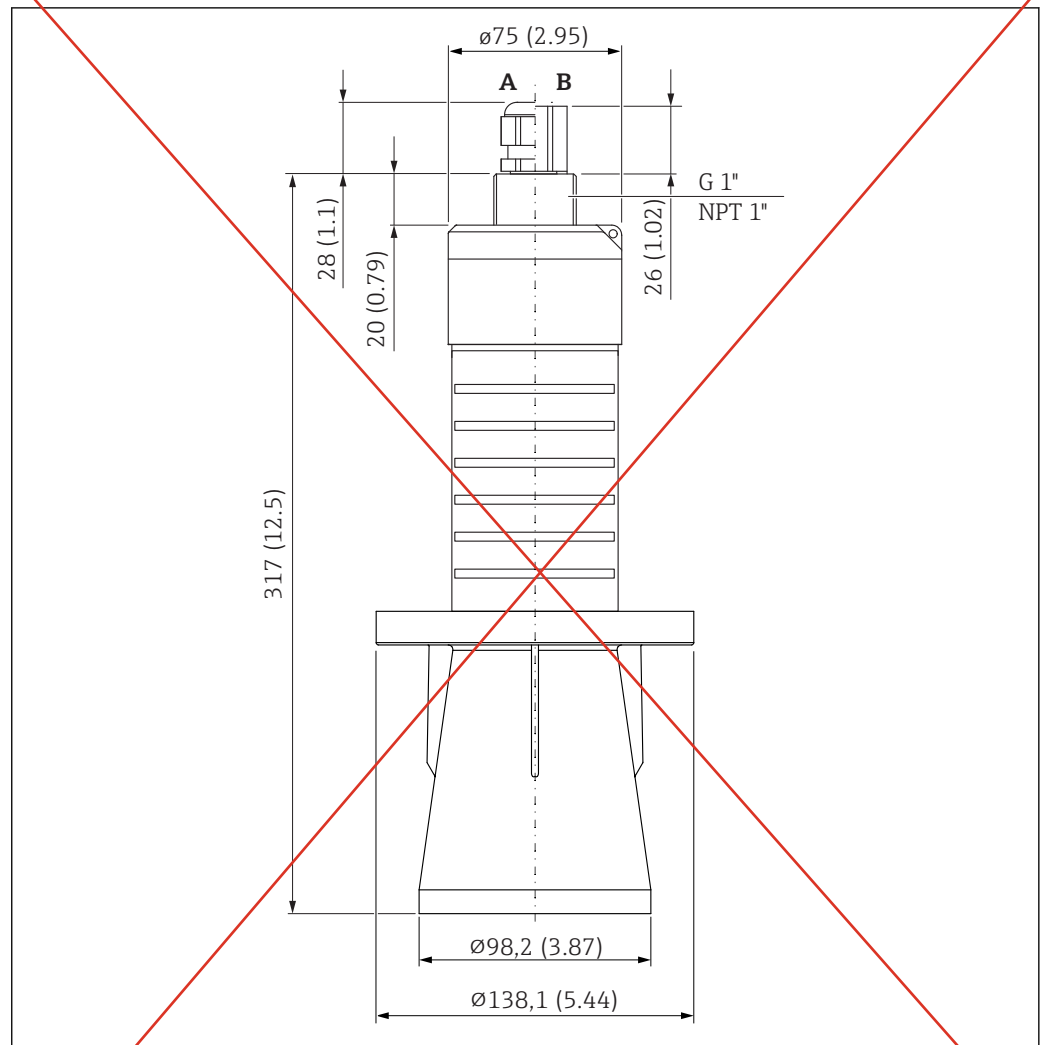
80 mm (3 in)Antenna



26 Dimensions of 80 mm (3 in) antenna; engineering unit: mm (in)

- A Cable gland
- B FNPT ½" conduit

80 mm (3 in) antenna with flooding protection tube

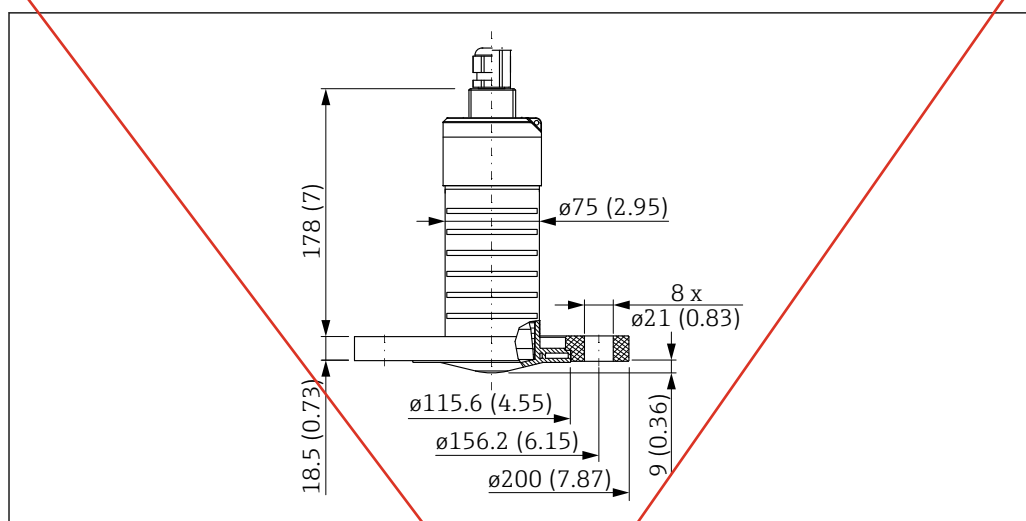


27 Dimensions of 80 mm (3 in) antenna mounted with flooding protection tube, engineering unit: mm (in)

A Cable gland

B FNPT 1/2" conduit

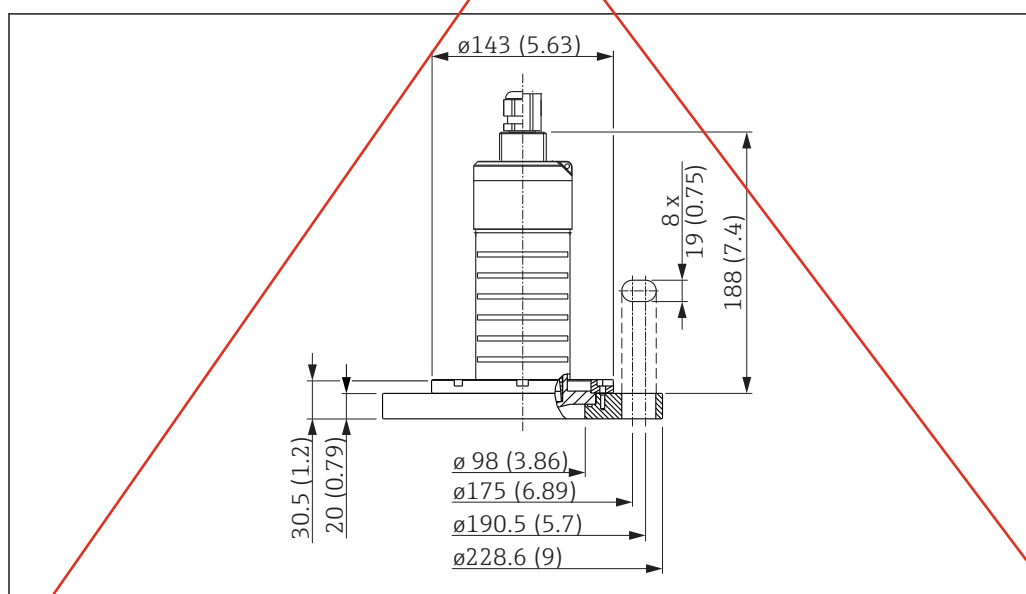
The flooding protection tube, metalized PBT-PC, can be ordered together with the device via the product structure "Accessory enclosed".

80 mm (3 in) antenna with slip-on flange 3"/DN80

A0028813

28 Dimensions of 80 mm (3 in) antenna with slip-on flange 3"/DN80, engineering unit: mm (in)

The slip-on flange 3"/DN80, PVDF, can be ordered together with the device via the product structure "Accessory enclosed".

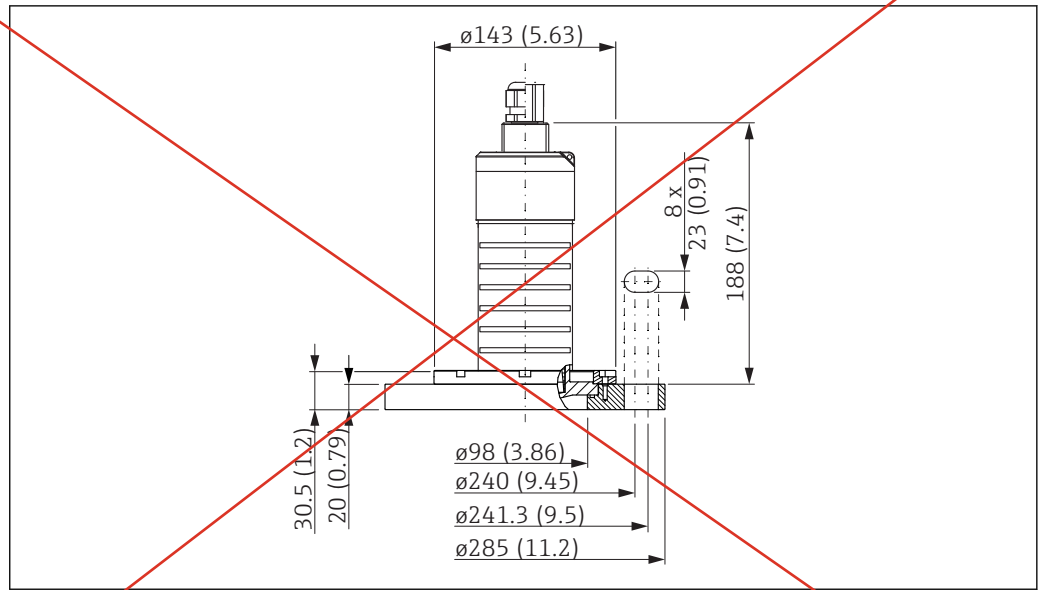
80 mm (3 in) antenna with slip-on flange 4"/DN100

A0028816

29 Dimensions of 80 mm (3 in) antenna with slip-on flange 4"/DN100, engineering unit: mm (in)

The slip-on flange 4"/DN100, PVDF, can be ordered together with the device via the product structure "Accessory enclosed".

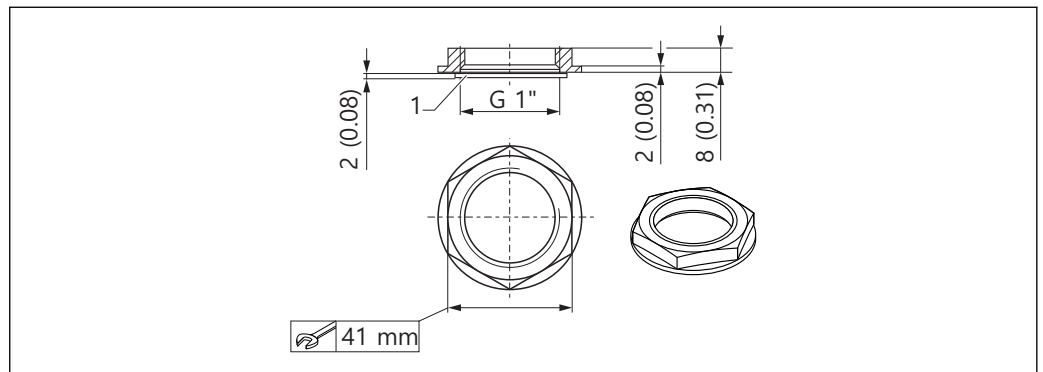
80 mm (3 in) antenna with slip-on flange 6"/DN150



30 Dimensions of 80 mm (3 in) antenna with slip-on flange 6"/DN150, engineering unit: mm (in)

The slip-on flange 6"/DN150, PVDF, can be ordered together with the device via the product structure "Accessory enclosed".

Counter nut for process connection, rear side



31 Dimensions of counter nut for process connection, rear side, engineering unit: mm (in)

1 Seal

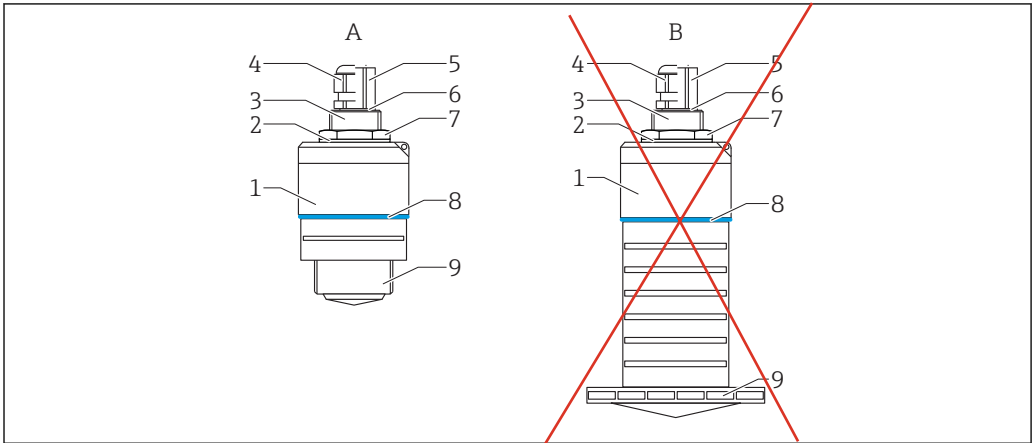
- The counter nut with seal (EPDM) is included in the scope of supply.
- Material: PA66

Weight

Weight (incl. 5 m (16.4 ft) cable)

- Device with 40 mm (1.5 in) antenna: approx. 2.5 kg (5.5 lb)
- Device with 80 mm (3 in) antenna: approx. 2.8 kg (6.2 lb)

Materials



A0028416

32 Overview of materials

- A 40 mm (1.5 in) Antenna
- ~~B 80 mm (3 in) Antenna~~
- 1 Sensor housing; PVDF
- 2 Seal; EPDM
- 3 Process connection, rear side; PVDF
- 4 Cable gland; PA
- 5 Conduit adapter; CuZn nickel-plated
- 6 O-ring; EPDM
- 7 Counter nut; PA6.6
- 8 Design ring; PBT-PC
- 9 Process connection, front side; PVDF

Connecting cable

Available cable length: 5 to 300 m (16 to 980 ft)

Material : PVC

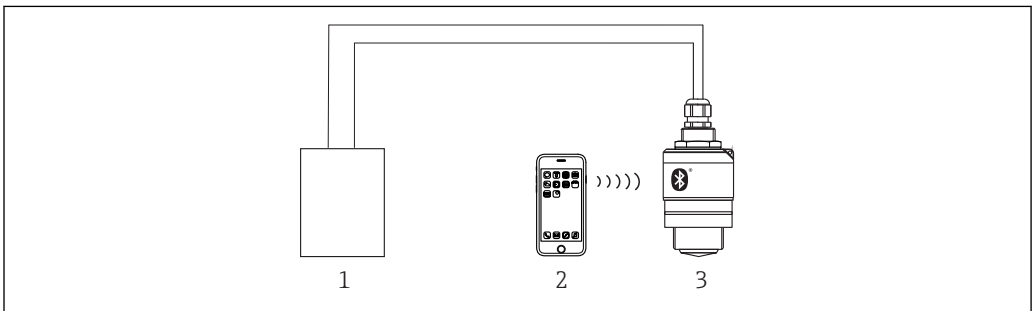
→ 20m

Operability

Operating concept

- 4 to 20 mA, HART
- Menu guidance with brief explanations of the individual parameter functions in the operating tool
- Optional: SmartBlue (app) via Bluetooth® wireless technology

Operation via Bluetooth® wireless technology

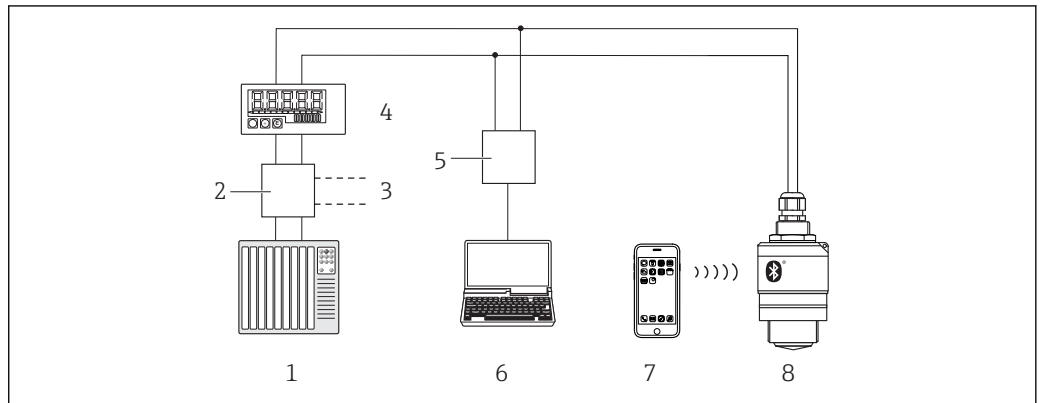


A0028895

33 Possibilities for remote operation via Bluetooth® wireless technology

- 1 Transmitter power supply unit
- 2 Smartphone/tablet with SmartBlue (app)
- 3 Transmitter with Bluetooth® wireless technology

Via HART protocol



A0028894

34 Options for remote operation via HART protocol

- 1 PLC (programmable logic controller)
- 2 Transmitter power supply unit, e.g. RN221N (with communication resistor)
- 3 Connection for Commubox FXA195
- 4 Loop-powered RIA15 process indicator
- 5 Commubox FXA195 (USB)
- 6 Computer with operating tool (FieldCare, DeviceCare)
- 7 Smartphone / tablet with SmartBlue (app)
- 8 Transmitter with Bluetooth® wireless technology

Certificates and approvals



The availability of approvals and certificates can be called up daily via the Product Configurator.

CE mark	The measuring system meets the legal requirements of the applicable EU Directives. These are listed in the corresponding EU Declaration of Conformity along with the standards applied. Endress+Hauser confirms successful testing of the device by affixing to it the CE mark.
RoHS	The measuring system complies with the substance restrictions of the Restriction on Hazardous Substances Directive 2011/65/EU (RoHS 2).
EAC conformity	The measuring system meets the legal requirements of the applicable EAC guidelines. These are listed in the corresponding EAC Declaration of Conformity together with the standards applied. Endress+Hauser confirms successful testing of the device by affixing to it the EAC mark.
RCM-Tick marking	The supplied product or measuring system meets the ACMA (Australian Communications and Media Authority) requirements for network integrity, interoperability, performance characteristics as well as health and safety regulations. Here, especially the regulatory arrangements for electromagnetic compatibility are met. The products are labelled with the RCM- Tick marking on the name plate.



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Approvals	■ Non-hazardous area ■ ATEX II 1 G Ex ia IIC T4 Ga ■ ATEX II 1/2 G Ex ia IIC T4 Ga/Gb ■ CSA C/US General Purpose ■ CSA C/US IS Cl.I Div.1 Gr.A-D, AEx ia / Ex ia T4 ■ CSA C/US Cl.I Div.2 Gr.A-D, T4 ■ EAC Ex ia IIC T4 Ga/Gb ■ IEC Ex ia IIC T4 Ga/Gb ■ KC Ex ia IIC T4 Ga/Gb ³⁾ ■ INMETRO Ex ia IIC T4 Ga/Gb ■ NEPSI Ex ia IIC T4 Ga/Gb ■ TIIS Ex ia IIC T4 ³⁾ Additional safety instructions must be followed for use in hazardous areas. Please refer to the separate "Safety Instructions" (XA) document included in the delivery. Reference to the applicable XA can be found on the nameplate.
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Explosion-protected smartphones and tablets	Only mobile end devices with Ex approval may be used in hazardous areas.
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Pressure equipment with allowable pressure ≤ 200 bar (2 900 psi)	Pressure instruments with a flange and threaded boss that do not have a pressurized housing do not fall within the scope of the Pressure Equipment Directive, irrespective of the maximum allowable pressure. Reasons: According to Article 2, point 5 of EU Directive 2014/68/EU, pressure accessories are defined as "devices with an operational function and having pressure-bearing housings". If a pressure instrument does not have a pressure-bearing housing (no identifiable pressure chamber of its own), there is no pressure accessory present within the meaning of the Directive.
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³⁾ Under development at time of going to press

harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes: (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

[Any] Changes or modifications made to this equipment not expressly approved by Endress+Hauser may void the FCC authorization to operate this equipment.

i This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna
- Increase the separation between the equipment and receiver
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected
- Consult the dealer or an experienced radio/TV technician for help

i

- The installation of the LPR/TLPR device shall be done by trained installers, in strict compliance with the manufacturer's instructions.
- The use of this device is on a "no-interference, no-protection" basis. That is, the user shall accept operations of high-powered radar in the same frequency band which may interfere with or damage this device. However, devices found to interfere with primary licensing operations will be required to be removed at the user's expense.
- Only for usage without the accessory "flooding protection tube", i.e. NOT in the free-field: This device shall be installed and operated in a completely enclosed container to prevent RF emissions, which can otherwise interfere with aeronautical navigation.

FCC / Industry Canada IDs

Tank level-probing radar

- **HVIN: FMR20**
 - FCC ID: LCGFMR2XK
 - Industry Canada ID: 2519A-2K
- **HVIN: FMR20X**
 - FCC ID: LCGFMR2XKT
 - Industry Canada ID: 2519A-2KT

Level-probing radar:

- **HVIN: FMR20+R7; FMR20+R8**
 - FCC ID: LCGFMR2XKF
 - Industry Canada ID: 2519A-2KF
- **HVIN: FMR20+R7X; FMR20+R8X**
 - FCC ID: LCGFMR2XKL
 - Industry Canada ID: 2519A-2KL

Japanese Radio Law and Japanese Telecommunications Business Law Compliance

This device is granted pursuant to the Japanese Radio Law (電波法) and the Japanese Telecommunications Business Law (電気通信事業法). This device should not be modified (otherwise the granted designation number will become invalid).

Certified No.: 202-LSF004

The products are labelled with the Technical Conformity Mark (GITEKI) from Japanese Ministry of Internal Affairs and Communications (MIC) on the name plate.



R 202-LSF004

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Mexico

El funcionamiento de este equipo está sujeto a las dos condiciones siguientes:

- (1) Este equipo o aparato no puede causar interferencias perjudiciales.
- (2) Este equipo o aparato debe aceptar todas las interferencias, incluyendo las que puedan causar un funcionamiento indeseado del equipo o aparato.

Este producto contiene un módulo inalámbrico

Marca: Endress+Hauser

Modelo: FMR20



A0034100

Other standards and guidelines

- IEC/EN 61010-1
Protection Measures for Electrical Equipment for Measurement, Control, Regulation and Laboratory Procedures
- IEC/EN 55011
"EMC Emission, RF Emission for Class B". Industrial, scientific and medical equipment – Electromagnetic disturbance characteristics - Limits and methods of measurement
- IEC/EN 61000-4-2
EMC Immunity, ESD (Performance Criteria A). Electromagnetic compatibility (EMC): Testing and measurement techniques - Electrostatic discharge immunity test (ESD)
- IEC/EN 61000-4-3
EMC Immunity, RF field susceptibility (Performance Criteria A). Electromagnetic compatibility (EMC): Testing and measurement techniques - Radiated, radio-frequency, electromagnetic field immunity test
- IEC/EN 61000-4-4
EMC Immunity, bursts (Performance Criteria B). Electromagnetic compatibility (EMC): Testing and measurement techniques - Electrical fast transient/burst immunity test
- IEC/EN 61000-4-5
EMC Immunity, surge (Performance Criteria B). Electromagnetic compatibility (EMC): Testing and measurement techniques - Surge immunity test
- IEC/EN 61000-4-6
EMC Immunity, conducted RF (Performance Criteria A). Electromagnetic compatibility (EMC): Testing and measurement techniques - Immunity to conducted disturbances induced by radio-frequency fields
- IEC/EN 61000-4-8
EMC Immunity, magnetic fields 50 Hz. Electromagnetic compatibility (EMC): Testing and measurement techniques - Power frequency magnetic field immunity test
- EN 61000-6-3
EMC Emission, conducted RF. EMC: Radiated interference - Residential, commercial and light industry environment
- NAMUR NE 21
Electromagnetic compatibility (EMC) of industrial process and laboratory control equipment
- NAMUR NE 43
Standardization of the signal level for the breakdown information of digital transmitters with analog output signal.
- NAMUR NE 107
Status classification as per NE107
- NAMUR NE 131
Requirements for field devices for standard applications
- IEEE 802.15.1
Requirements for the *Bluetooth*® wireless technology interface

Ordering information

Detailed ordering information is available for your nearest sales organization www.addresses.endress.com or in the Product Configurator under www.endress.com :

1. Click Corporate
2. Select the country
3. Click Products
4. Select the product using the filters and search field
5. Open the product page

The Configuration button to the right of the product image opens the Product Configurator.

Product Configurator - the tool for individual product configuration

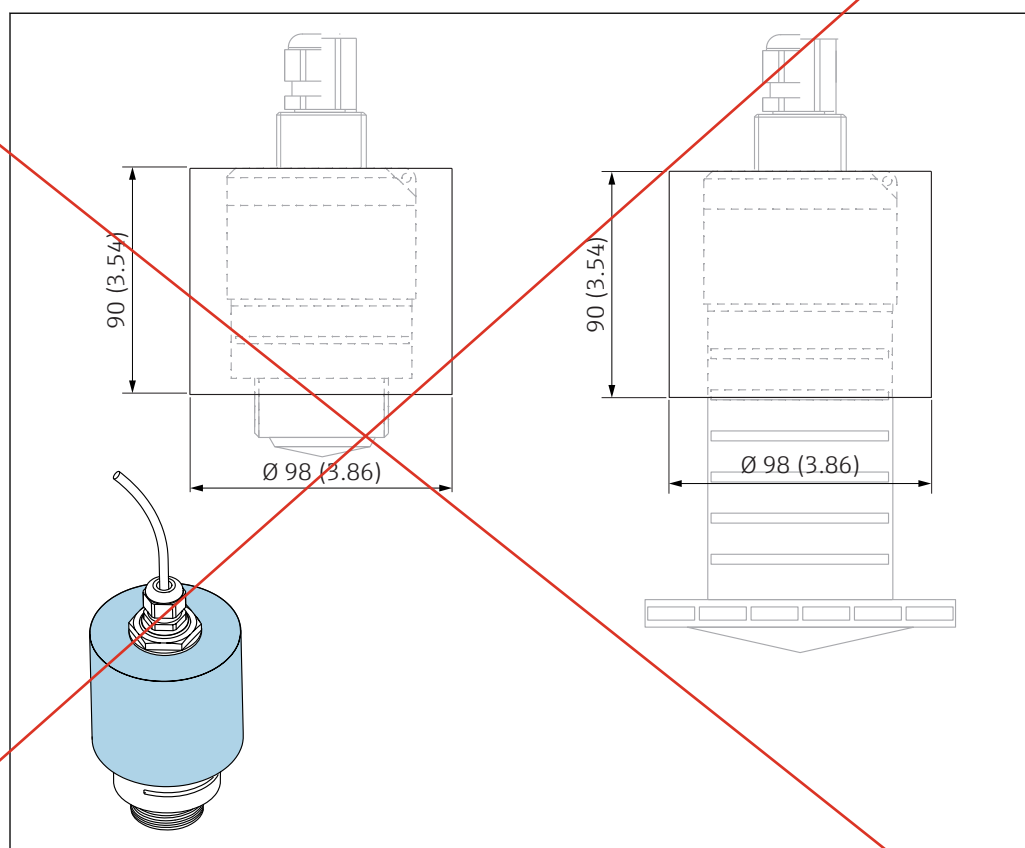
- Up-to-the-minute configuration data
- Depending on the device: Direct input of measuring point-specific information such as measuring range or operating language
- Automatic verification of exclusion criteria
- Automatic creation of the order code and its breakdown in PDF or Excel output format
- Ability to order directly in the Endress+Hauser Online Shop

Accessories

Device-specific accessories

Protective cover

The protective cover can be ordered together with the device via the product structure "Accessory enclosed".



 35 Dimensions of protective cover, engineering unit: mm (in)

A0028841

Material
PVDF